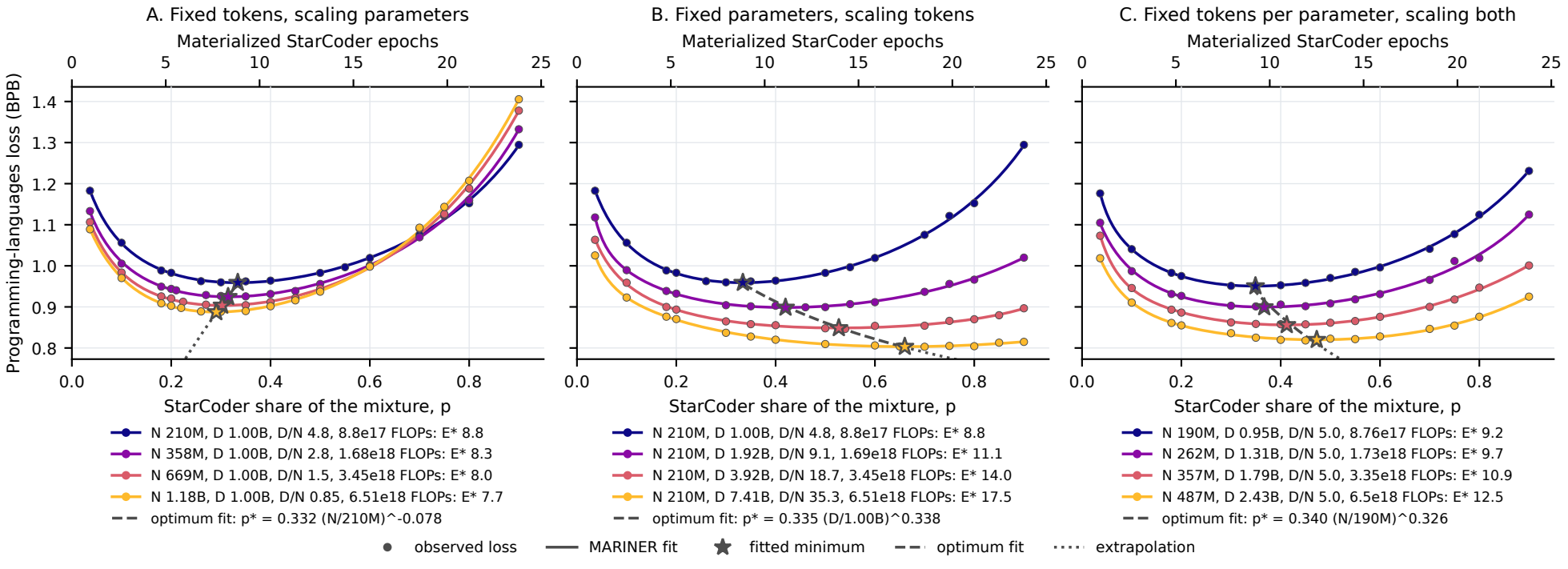
A. Fixed tokens, scaling parameters
B. Fixed parameters, scaling tokens
C. Fixed tokens per parameter, scaling both
Materialized StarCoder epochs
Programming-languages loss (BPB)
StarCoder share of the mixture, p
N 210M, D 1.00B, D/N 4.8, 8.8e17 FLOPs: E* 8.8
N 358M, D 1.00B, D/N 2.8, 1.68e18 FLOPs: E* 8.3
N 669M, D 1.00B, D/N 1.5, 3.45e18 FLOPs: E* 8.0
N 1.18B, D 1.00B, D/N 0.85, 6.51e18 FLOPs: E* 7.7
optimum fit: p* = 0.332 (N/210M)^-0.078
N 210M, D 1.00B, D/N 4.8, 8.8e17 FLOPs: E* 8.8
N 210M, D 1.92B, D/N 9.1, 1.69e18 FLOPs: E* 11.1
N 210M, D 3.92B, D/N 18.7, 3.45e18 FLOPs: E* 14.0
N 210M, D 7.41B, D/N 35.3, 6.51e18 FLOPs: E* 17.5
optimum fit: p* = 0.335 (D/1.00B)^0.338
N 190M, D 0.95B, D/N 5.0, 8.76e17 FLOPs: E* 9.2
N 262M, D 1.31B, D/N 5.0, 1.73e18 FLOPs: E* 9.7
N 357M, D 1.79B, D/N 5.0, 3.35e18 FLOPs: E* 10.9
N 487M, D 2.43B, D/N 5.0, 6.5e18 FLOPs: E* 12.5
optimum fit: p* = 0.340 (N/190M)^0.326
observed loss
MARINER fit
fitted minimum
optimum fit
extrapolation